Planet Formation through the Lens of Dynamics

Proefschrift

ter verkrijging van
de graad van doctor aan de Universiteit Leiden,
op gezag van rector magnificus prof.dr. S. de Rijcke,
volgens besluit van het college voor promoties
te verdedigen op vrijdag 19 juni 2026
klokke 11:30 uur

door

Shuo Huang

geboren te Hefei, China
in 1998

Promotiecommissie

Promotores:	Prof. dr. Simon Portegies Zwart	Leiden University
	Prof. dr. Chris Ormel	Tsinghua University
Overige leden:	Prof. dr. xx	xx

ISBN: xx

Front cover:
Something for the cover.

We all learn at different stages and speeds, and each of us has our own strengths and areas of expertise.

– Han Yu (c. 768–824), Tang dynasty

Contents

1 Introduction **1**
1.1 Planet formation . . . 2
1.1.1 Protoplanet disk . . . 3
1.1.2 Planet migration . . . 4
1.2 Mean motion resonance . . . 5
1.3 Planet architecture . . . 8
1.3.1 Exoplanet systems . . . 8
1.3.2 The Solar System . . . 9
1.4 This Thesis . . . 10
1.5 Outlook: confirm and characterize broken chains . . . 11

2 When, where, and how many exoplanets end up in orbital resonances? **15**
2.1 Introduction . . . 16
2.2 Methodology . . . 17
2.2.1 Disc model . . . 18
2.2.2 Type I migration . . . 19
2.2.3 Dynamics of resonance trapping . . . 19
2.2.4 Mass-radius relations . . . 20
2.2.5 A statistical model of resonant and non-resonant planets . . . 22
2.3 Resonance Trapping criterion for the restricted 3-body problem . . . 24
2.3.1 Theoretical Derivation . . . 24
2.3.2 Comparison with simulation . . . 25
2.4 MCMC analysis . . . 27
2.4.1 Sample selection . . . 27
2.4.2 Implication on planet-disc interaction from MCMC . . . 29
2.5 Implications for planet formation . . . 32
2.5.1 Fraction of resonant pairs . . . 32
2.5.2 Upper limit on the disc surface density at resonance trapping . . . 34
2.5.3 Migration barrier reflects the disc inner rim . . . 36
2.6 Discussion . . . 38
2.7 Conclusions . . . 41
Appendices . . . 41
2.A Results of the General disc model . . . 41
2.B MCMC performance examination . . . 42
2.B.1 Tests without assuming a disc structure . . . 42
2.B.2 Tests assuming a disc structure . . . 43

3 The dynamics of the special resonance chain systems and their formation: TRAPPIST-1 49
3.1 Introduction . . . 50
3.2 Model . . . 52
3.2.1 Preliminaries and Terminology . . . 52
3.2.2 Numerical integration . . . 54
3.2.3 Planet-disc interaction . . . 55
3.2.4 Tidal dissipation . . . 58
3.3 Preliminary simulations . . . 59
3.3.1 Planet b at the cavity edge (bC) . . . 59
3.3.2 Formation imprints for planet c and g (bCcgI) . . . 61
3.3.3 Planet d at the cavity edge (dCcgI) . . . 62
3.3.4 Parameter variation for Objective I (dCcgI) . . . 64
3.4 Trapping and escape of planets in 3BRs . . . 65
3.4.1 Crossing and trapping of the three-body resonances (dCcgIcT) . . . 67
3.4.2 Long term tidal dissipation (dCcgIcT) . . . 70
3.5 Early cavity infall (dCcgIcL) . . . 70
3.5.1 Disc repulsion on planets in the cavity . . . 71
3.5.2 Parameter study (dCcgIcL) . . . 72
3.6 Analysis . . . 74
3.6.1 Trapping under convergent migration in MMRs . . . 74
3.6.2 Eccentricity-period relation in 3BRs . . . 75
3.7 Discussion . . . 76
3.7.1 Model Assessment . . . 76
3.7.2 Resonances Establishing and Breaking . . . 77
3.7.3 Application to other systems . . . 78
3.8 Conclusions . . . 79
Appendices . . . 80
3.A Lindblad torque . . . 80

4 The dynamics of the broken resonance chain systems and their formation: the Solar System 81
4.1 Introduction and motivation . . . 82
4.2 Model . . . 84
4.2.1 Initial conditions . . . 84
4.2.2 Giant planet instability . . . 85
4.2.3 Theia and proto-Earth mass-radius calculation . . . 85
4.3 Comparison with Solar system . . . 87
4.4 Implications . . . 90
4.5 Conclusions . . . 91
Appendices . . . 92
4.A Formation of a resonant chain in the pebble accretion and planetesimal accretion models . . . 92
4.B Parameter sensitivity . . . 96
4.C Venus-Mars period ratio in the late collisional growth scenario . . . 98

5 Birth stellar cluster dynamics matters: planet population synthesis with external photo-evaporation 101
5.1 Introduction . . . 102
5.2 Methods . . . 103
5.2.1 Cluster evolution . . . 103
5.2.2 Disk model . . . 105
5.2.3 Planet growth . . . 106
5.2.4 Planet migration . . . 109
5.3 Initial conditions and parameter choices . . . 109
5.4 Results . . . 111
5.4.1 Planet formation for two example stars . . . 111
5.4.2 Planet population: Isolated versus clustered . . . 113
5.4.3 Co-evolution between CJ formation and the cluster . . . 115
5.5 Analytical criteria for CJ formation in the cluster . . . 115
5.6 Discussion . . . 118
5.6.1 Model assessment . . . 118
5.6.2 Cold Jupiter and Neptune in observation . . . 119
5.7 Conclusions . . . 120
Appendices . . . 121
5.A Planet population in isolated stars and a star cluster . . . 121

6 ALMA signature of closely-spaced pebble-accreting protoplanets in Transition disks 123
6.1 Introduction . . . 124
6.2 Methods . . . 125
6.2.1 Dust transport including pebble accretion . . . 125
6.2.2 Initial conditions . . . 126
6.3 Pebble flux reduction . . . 126
6.3.1 Pebble accretion efficiency . . . 126
6.3.2 Dust cavity and ring structure . . . 128
6.4 Two groups of planet-hosting transition disks . . . 130
6.4.1 Below pebble isolation mass: GM Aur . . . 132
6.4.2 Above pebble isolation mass: PDS 70 . . . 134
6.4.3 A sample of transition disks . . . 136
6.5 Discussion . . . 138
6.5.1 Model assessment . . . 138
6.5.2 Feasibility of the "sweet spot" on the accretion map . . . 139
6.5.3 Emerging the planet formation story in transition disks . . . 139
6.5.4 Consequence for planet detection . . . 140
6.6 Conclusions . . . 140
Appendices . . . 141
6.A Numerical treatment of pebble flux reduction . . . 141
6.B Evaluating dust diffusion during pebble accretion . . . 142
6.C Simulated continuum of GM Aur with three planets . . . 142
6.D Profile of the gap opened by planet . . . 143
6.E Quantify the morphology of dust rings . . . 144

Bibliography 145

Summary **165**

Nederlandse samenvatting **169**

List of Publications **173**

Curriculum Vitæ **175**

Acknowledgments **177**